

GST Invoice Generator - Simple Billing

1. Project Overview

Project Name: GST Invoice Generator (MVP)

Goal: Build a simple SaaS web app that allows small traders to:

- Create GST invoices
- Automatically calculate GST
- Save invoices
- Download & print PDF invoices

Target Users:

non-technical small traders and service providers.

2. Scope

- User login (email + password)
- Forgot password via email reset link
- Create GST invoice
- Auto GST calculation (CGST/SGST or IGST)
- Save invoice
- View invoice list (dashboard)
- Download & print PDF invoice
- Optional business logo upload (single logo per user)
- Optional business email and/or phone in invoice header
- Deployed live URL

3. Tech Stack

Frontend

- React (Vite)
- Tailwind CSS
- React Hook Form

Backend

- Supabase
 - Authentication
 - Postgres Database
 - Row Level Security (RLS)
 - Storage (for logo upload)

Authentication

- Email + Password
- Forgot password → email reset link

PDF Generation

- Client-side HTML → PDF
- Library: html2pdf.js

Hosting

- Vercel

4. Core Features (Detailed)

4.1 Authentication

- User registers with:
 - Email
 - Password
- User logs in using email + password
- Forgot password sends reset email

4.2 Business Settings (User Profile)

User can save:

- Business Name (required)
- State (required)
- GSTIN (optional)
- Business Email (optional)
- Business Phone (optional)
- Business Logo (optional)

Logo Rules:

- One logo per user
- PNG / JPG / SVG only
- Max size: 500 KB
- Stored in Supabase Storage
- Used only in invoice PDF header
- If no logo → fallback to business name text

4.3 Invoice Creation

Invoice must include:

Seller

- Business Name
- State
- GSTIN (optional)

- Email and/or Phone (optional, from settings)
- Logo (optional)

Buyer

- Name
- State
- GSTIN (optional)

Invoice Info

- Invoice Number (auto-generated)
- Date (default: today)

Items (max 20)

- Description
- Quantity
- Unit Price
- GST Rate (0%, 5%, 12%, 18%, 28%)

4.4 GST Calculation Logic (IMPORTANT)

If seller State = buyer State:

CGST = GST / 2

SGST = GST / 2

IGST = 0

Else:

IGST = GST

CGST = 0

SGST = 0

- All calculations done in frontend
- Round all values to 2 decimal places

4.5 Dashboard

- List all saved invoices
- Show:
 - Invoice Number
 - Date
 - Total Amount
- Actions:
 - View
 - Download PDF

4.6 PDF Generation

PDF Requirements

- A4 size
- Simple, professional layout
- Clean and readable

Header must include:

- Business Name (always)
- Logo (optional)
- GSTIN (if available)
- Email and/or Phone (if available)

PDF Content Includes:

- Seller details
- Buyer details
- Item table
- GST breakup (CGST/SGST or IGST)
- Grand total

File Name Format:

BusinessName_InvoiceAmount_InvoiceNo_Date.pdf

5. Database Structure (Minimal)

users

- id (uuid)
- email
- business_name
- state
- gstin
- business_email (optional)
- business_phone (optional)
- logo_url (optional)
- created_at

invoices

- id
- user_id
- invoice_no
- invoice_data (JSON)
- total_amount
- created_at

6. UI Guidelines

- Medium to large buttons
- Simple, readable text
- Mobile-friendly layout
- No animations
- No dark mode
- No fancy fonts

Success Message Example: “Invoice saved successfully”

Error Message Example: “Please enter item price”

7. Definition of DONE (MVP)

The project is considered DONE only if:

- User can register and log in
- Business details can be saved
- Optional logo upload works
- Invoice can be created
- GST calculation is correct
- Invoice saves successfully
- PDF downloads and prints correctly
- App is deployed and accessible via live URL